

We live in what is called the Digital Age

- phones
- computers
- smart TVs
- digital camera

} These are all digital systems. A digital system has two definitions:

Simple definition: anything that has an embedded digital computer

an embedded computer is a collection of circuits
that perform a specific task.

complex definition: hardware used to process information in discrete form.

Information is processed using letters, numbers,
symbols, etc (separate, countable values)

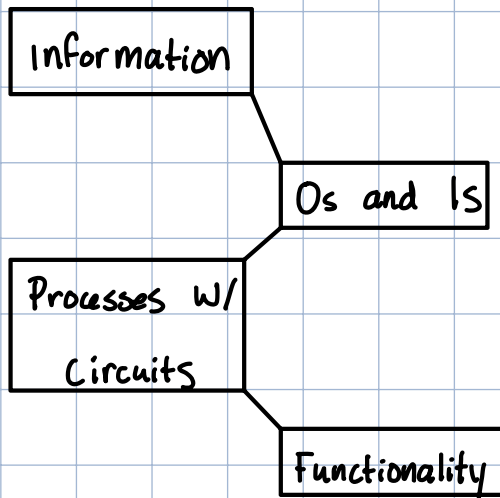
A digital computer has generality:

- a single computer can do different tasks because it follows specific instructions called programs
- Because instructions can be manipulated, hardware doesn't need to be remade.

Digital systems represent information using signals, a quantity that carries information (different states)

- electrical system: voltage (high\low)
- digital system: binary (0s\1s)

Digital System Flow:



A number system is a rule for writing numbers. It tells us which digits and symbols can be used to represent numbers.

We use the decimal number system: $\left\{ \begin{array}{l} (576)_{10} = 500 + 70 + 6 \\ \uparrow \\ \text{base 10} \quad (5 \cdot 100) + (7 \cdot 10) + (6 \cdot 1) \\ (5 \cdot \underline{10^2}) + (7 \cdot \underline{10^1}) + (6 \cdot \underline{10^0}) \end{array} \right.$

[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
10 digits allowed

There is also the octal number system: $\left\{ \begin{array}{l} (576)_8 = (5 \cdot \underline{8^2}) + (7 \cdot \underline{8^1}) + (6 \cdot \underline{8^0}) \\ \uparrow \\ \text{base 8} \end{array} \right.$

[0, 1, 2, 3, 4, 5, 6, 7]
8 digits allowed

Other number systems include base 5: $\left\{ \begin{array}{l} (4213)_5 = (4 \cdot 5^3) + (2 \cdot 5^2) + (1 \cdot 5^1) + (3 \cdot 5^0) \\ \uparrow \\ \text{base 5} \end{array} \right.$

[0, 1, 2, 3, 4]
5 digits allowed

Number systems can be written in the expanded form (coefficient · base power)

$$(a_n \dots a_1 a_0)_r = (a_n \cdot r^n) + \dots (a_1 \cdot r^1) + (a_0 \cdot r^0)$$

$\uparrow$ individual digit $\uparrow$ base

A binary number is a number in the base 2 system. Each binary number is composed of 0s and 1s.

$$\underbrace{[0, 1]}_{2 \text{ digits}} \left\{ \begin{array}{l} (101)_2 = (1 \cdot 2^2) + (0 \cdot 2^1) + (1 \cdot 2^0) \\ \quad \quad \quad \uparrow \\ \quad \quad \quad \text{base 2} \end{array} \right.$$

- Each individual digit is called a bit
- 4 bits = nibble $(1101)_2$
- 8 bits = byte $(01011101)_2$
- The farthest left bit is the most significant.

$$(101)_2 = (1 \cdot 2^2) + (0 \cdot 2^1) + (1 \cdot 2^0)$$

$$(101)_2 = (1 \cdot 4) + (0 \cdot 2) + (1 \cdot 1)$$

$$(101)_2 = (4) + (0) + (1)$$

$$(101)_2 = 5$$

$$(1001)_2 = (1 \cdot 2^3) + (0 \cdot 2^2) + (0 \cdot 2^1) + (1 \cdot 2^0)$$

$$(1001)_2 = (1 \cdot 8) + (0 \cdot 4) + (0 \cdot 2) + (1 \cdot 1)$$

$$(1001)_2 = (8) + (0) + (0) + (1)$$

$$(1001)_2 = 9$$

For every number in a non-decimal base, you can find its decimal equivalent by solving the expanded form. A shortcut would be to add the place values where 1 exists.

$$\dots \begin{array}{c|c|c|c} 1 & 0 & 0 & 1 \\ \hline 8 & 4 & 2 & 1 \\ 2^3 & 2^2 & 2^1 & 2^0 \end{array} = 8 + 1 = 9$$

Binary numbers can have radix points (decimal points)

$$(10.11)_2 = (1 \cdot 2^1) + (0 \cdot 2^0) + (1 \cdot 2^{-1}) + (1 \cdot 2^{-2})$$

$$(10.11)_2 = (1 \cdot 2) + (0 \cdot 1) + (1 \cdot 0.5) + (1 \cdot 0.25)$$

$$(10.11)_2 = (2) + (0) + (0.5) + (0.25)$$

$$(10.11)_2 = 2.75$$

Binary Addition Rules:

$$\begin{array}{r} 1) \ 0 \\ + 0 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 2) \ 1 \\ + 0 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 3) \ 0 \\ + 1 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 4) \ 1 \\ + 1 \\ \hline 10 \end{array}$$

$$\begin{array}{r} 5) \ 1 \\ + 1 \\ \hline 11 \end{array}$$

carry
overs

$$\begin{array}{r} \text{Ex: } 1010 \\ + 11 \\ \hline 1101 \end{array} \quad \begin{array}{r} 10 \\ + 3 \\ \hline 13 \end{array}$$

$$\begin{array}{r} \text{Ex: } 1011 \\ + 11 \\ \hline 1110 \end{array} \quad \begin{array}{r} 11 \\ + 3 \\ \hline 14 \end{array}$$

Binary Multiplication Rules:

$$1) \ 0 \cdot 0 = 0$$

$$2) \ 1 \cdot 0 = 0$$

$$3) \ 1 \cdot 1 = 1$$

$$\begin{array}{r} \text{Ex: } 1010 \\ \times 11 \\ \hline 1010 \\ + 10100 \\ \hline 11110 \end{array} \quad \Leftrightarrow \quad (10 \cdot 3) = 30$$

$$\begin{array}{r} \text{Ex: } 1110 \\ \times 11 \\ \hline 1110 \\ + 11100 \\ \hline 101010 \end{array} \quad \Leftrightarrow \quad (14 \cdot 3) = 42$$

Binary Subtraction Rules:

$$1) \ 0 - 0 = 0$$

$$2) \ 1 - 0 = 1$$

$$3) \ 1 - 1 = 0$$

$$4) \ 0 - 1 = \text{borrowing}$$

$$\begin{array}{r} \text{Ex: } 1010 \\ - 11 \\ \hline 0111 \end{array} \quad \Leftrightarrow \quad \begin{array}{r} 10 \\ - 3 \\ \hline 7 \end{array}$$

This subtraction process is not
computer efficient.

